

SSN COLLEGE OF ENGINEERING (Autonomous)
(Affiliated to Anna University, Chennai)
DEPARTMENT OF CSE

UCS 2511 NETWORK LABORATORY

A13: Simulation of Routing Protocol

Faculty In-charge: Dr.N.Sujaadeen, Assistant Professor / CSE

Batch: 2020-2024 Year: III Sem: V Sec: A

Learning outcome: Simulation of Routing protocol

Algorithm:

1. Create 12 nodes and the links between the nodes as
 - a. 0 → 8 1Mb 10 ms duplex link droptail
 - b. 1 → 10 1Mb 10 ms duplex link droptail
 - c. 0 → 9 1Mb 10 ms duplex link droptail
 - d. 9 → 11 1Mb 10 ms duplex link droptail
 - e. 10 → 11 1Mb 10 ms duplex link droptail
 - f. 11 → 5 1Mb 10 ms duplex link droptail
2. Align all nodes properly
3. Setup a UDP connections over 0 and 5, 1 and 5 with flow id, type, packet size, rate, random fields.
4. Set different colors for different flows.
5. Use ***distance vector routing*** protocol.
6. Make links 11-5 and 7-6 down for 1 second.
7. Run the simulation for 5 seconds, and show the simulation in network animator and in trace file.

- Similarly write another tcl script to simulate ***link state routing*** protocol.

Also show the flooding in the simulation.